What we did and what we found

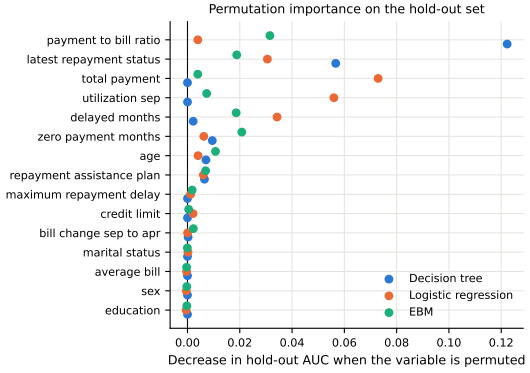
Task. Predict next-month payment difficulty (21 % positives; 25k labelled clients, 15 predictors).

Protocol. 80/20 development/hold-out split. *Every* choice (hyperparameters, log transform, threshold) is made with the *same* 5 CV folds; the hold-out set is used once for confirmation.

- Tuning by CV AUC; one-SE rule ⇒ simplest model
- No class weights: CV-chosen thresholds $t_{\text{acc}} \approx 0.5$ (submission), $t_{\text{bal}} \approx 0.2$ (analysis)
- Common yardsticks: permutation importance, log-odds effect curves, additive local decompositions

Findings. EBM best in 5/5 folds, and better than a black box. Same drivers in all models, but different *carriers* and *shapes* ⇒ local conflicts. **Submitted: EBM.**

5-fold CV	AUC	Acc.	Bal. acc.
Tree (17 leaves)	.752	.812	.692
Logistic regression	.761	.814	.695
EBM (`interactions=0`)	**.776**	**.817**	**.705**
Majority baseline	.500	.788	.500
Black box (HistGB)	.768	.815	.701



Decision tree: 17 leaves collapse into a 6-rule risk profile

Flag difficulty if any rule fires (status = latest repayment status, ratio = payment-to-bill ratio):

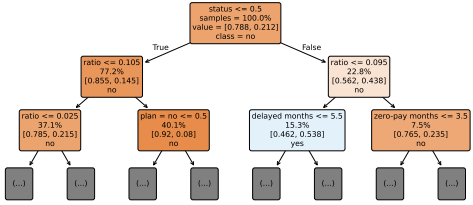
	Rule	Risk
R1	status ≥ 1 and ratio ≤ 0.095	.54
R2	status ≤ 0 and ratio ≤ 0.025	.43
R3	status ≥ 1, ratio > 0.095, zero-pay months ≥ 4	.33
R4	status ≤ 0, ratio $\in (0.025, 0.105]$, age ≤ 24	.32
R5	status ≥ 1, ratio $\in (0.095, 0.315]$, zero-pay ≤ 3	.28
R6	status ≤ 0, ratio $\in (0.025, 0.105]$, age ≥ 25, plan	.26

- R1 alone: 15 % of clients, 39 % of all difficulties
- Uses only 7 of 15 variables; 85 % of the impurity reduction comes from status and ratio
- Most *simulatable* model, but the coarsest (17 probability levels)

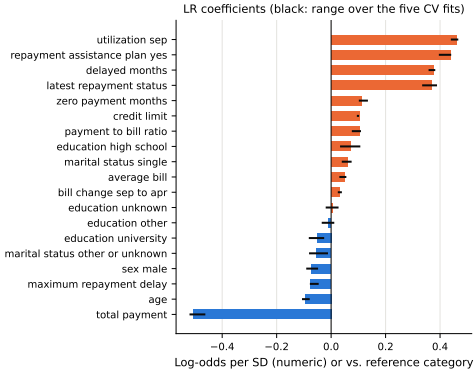
Local case C (actual: difficulty): 1 month late, 4/6 months delayed, plan, *negative bill* $\Rightarrow$ ratio 4.39.

Tree path: status $\geq 1 \rightarrow$ ratio $> 0.095 \rightarrow$ treated as *protective* (-1.25 log-odds) $\Rightarrow p = 0.15$, **missed**.

LR ($p = 0.50$) and EBM ($p = 0.55$) read the high ratio as *risk* and add the delayed months $\Rightarrow$ correctly flagged.



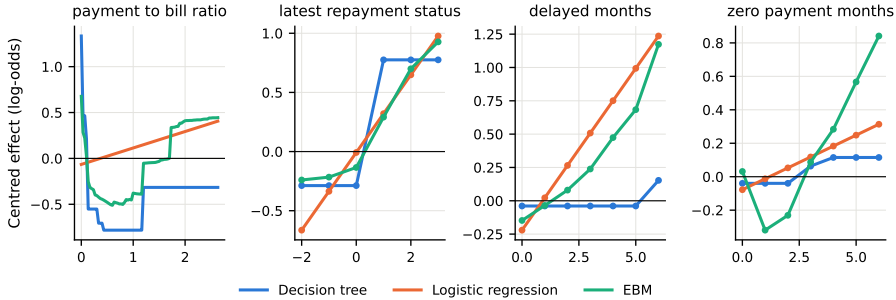
Logistic regression: compact, stable, but conditional and linear



- Standardized inputs, reference coding; a CV-selected signed-log transform of money variables raises AUC from .750 to .762
- Strongest effects: utilization $+0.46$/SD, assistance plan $+0.44$ (OR 1.55), delayed months $+0.38$, status $+0.37$, log total payment -0.51
- Signs stable over the 5 CV refits, **but conditional**: max. delay ($r = +.27$ with the target, $\beta = -.08$), credit limit and payment ratio all *reverse* their marginal sign because of collinearity
- Linear in the undocumented status codes $-2, -1, 0$

Local case D (actual: no difficulty): up to date on everything, but utilization 2.0 $\Rightarrow +1.80$ log-odds, $p = 0.39$, **false alarm** (EBM 0.28 also). The tree ignores utilization: $p = 0.06$, correct.

EBM: the shapes behind the best model



- **Payment ratio is U-shaped**: risk is lowest at 0.5–1.2 and rises again above 1.7 (catch-up payments, tiny bills). The tree is flat there; LR can only fit a straight line.
- **Zero-payment months**: 1–2 are protective, 3+ are harmful; **age** is U-shaped (under 25, over 57).
- Assistance plan +0.41: a marker of prior stress, *not* a causal harm.
- Permutation-importance rank correlation: tree–EBM .85, LR–EBM .73, tree–LR .36. This is the *Rashomon effect* on redundant payment variables.
- Local explanations are exact sums of the shapes. Case B (all correct): status 2 and the plan dominate in all three models.